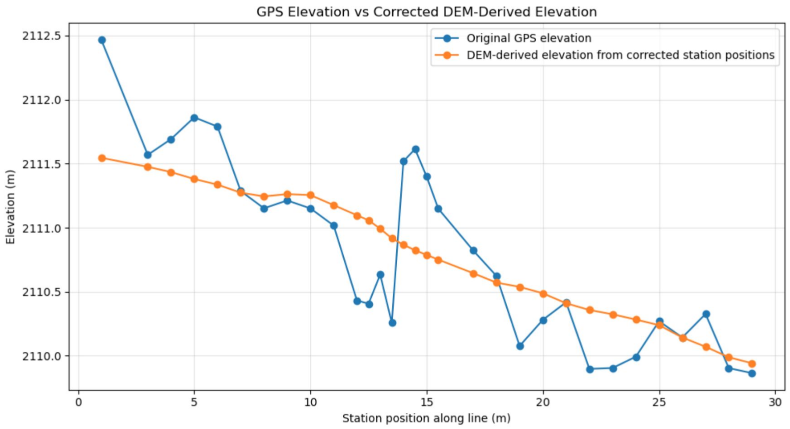
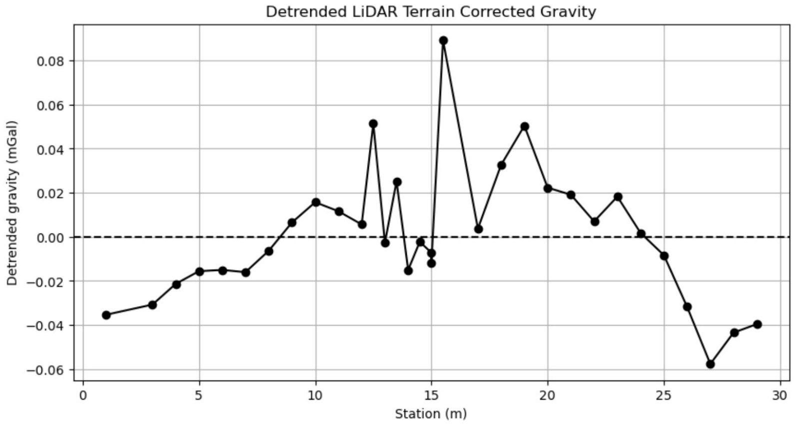

THE LORAIN COAL & DOCK COMPANY
The Miners
LORADO - COAL MINERS - APRIL 1918

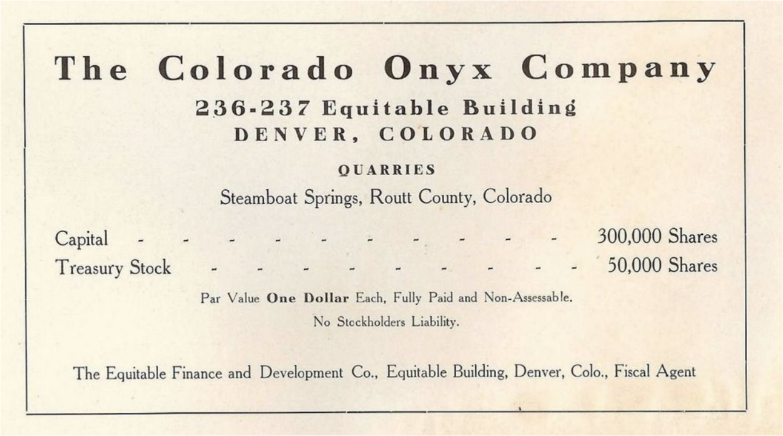
MORE gravity Hmmmmmm

- We found out that our elevations weren't making sense when we plotted them for final figures… Oopsie
- We derived new elevations from DEM data and resulted in a smooth trend
- BUT now we have no negative residual gravity anomaly… a positive anomaly?





New Literature



The Colorado Onyx Company

236-237 Equitable Building
DENVER, COLORADO

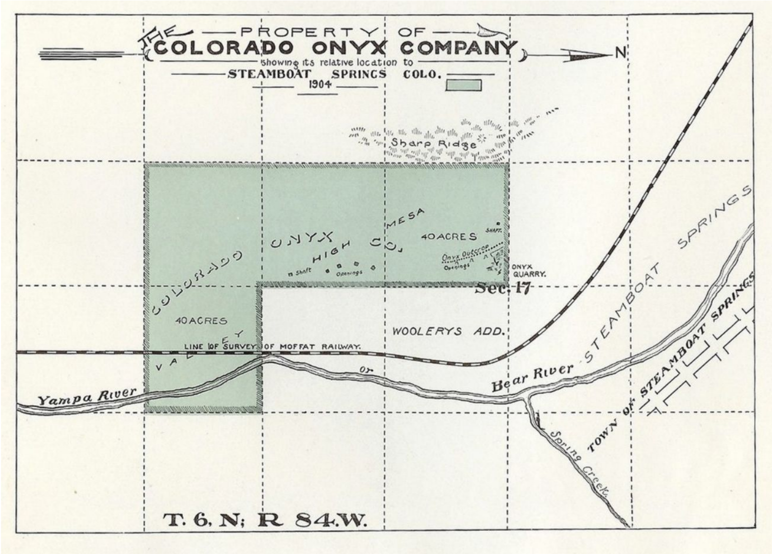
QUARRIES

Steamboat Springs, Routt County, Colorado

Capital	300,000 Shares
Treasury Stock	50,000 Shares

Par Value **One Dollar** Each, Fully Paid and Non-Assessable.
No Stockholders Liability.

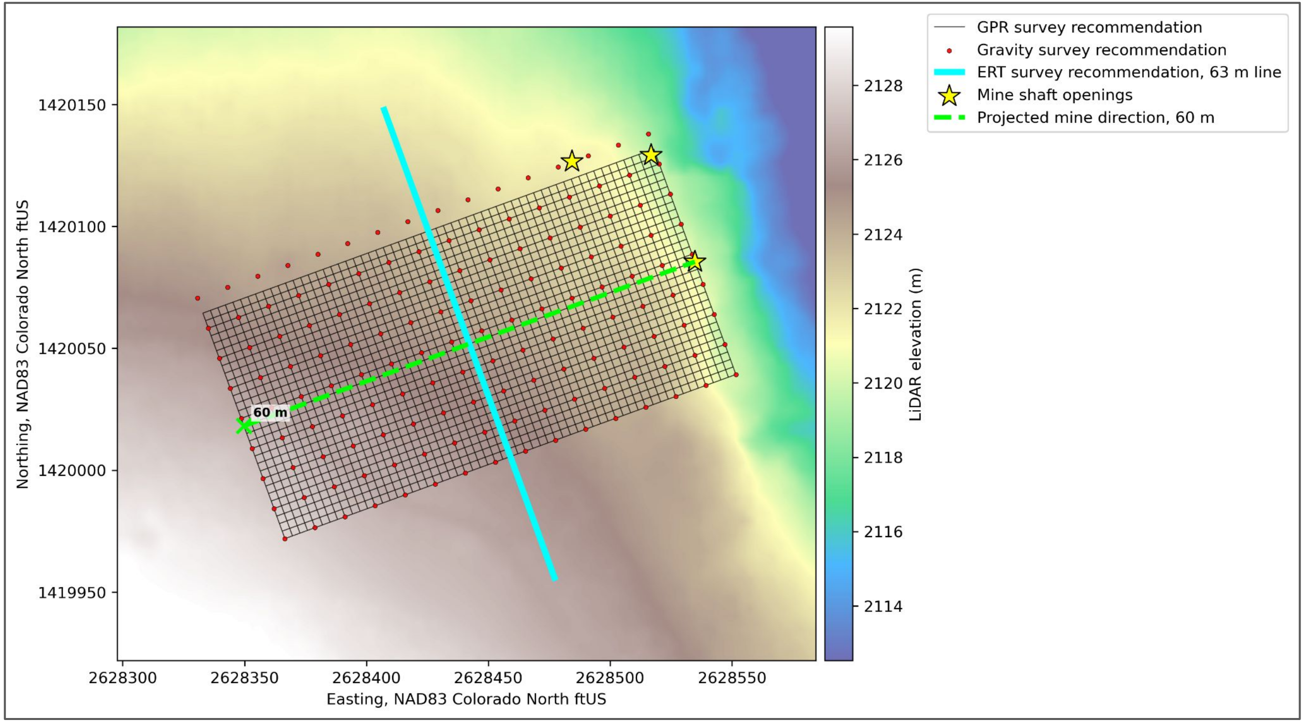
The Equitable Finance and Development Co., Equitable Building, Denver, Colo., Fiscal Agent



- Historical maps place the quarry claims on the hillside above the Yampa River.
- Literature suggests extraction followed travertine/onyx-bearing zones rather than a single vertical shaft.

Recommended surrey parameters in the future

- GPR survey is projected to take 1-2 hours
- Gravity survey is projected to take 5-6 hours with 2 gravimeters
- 1m spacing gpr
- 4m spacing gravity



Extended Abstract

It’s coming along!

Future Work

To improve upon this survey’s results we recommend future teams to revisit the onyx mine and conduct a 3 method survey utilizing gravity, GPR, and ERT.

Gravity:

Teams should construct a 60 m x 30 m grid with lines extending NW-SE perpendicular to lines extending NE-SW atop the Onyx Mine; 4m spacing between gravimeters along these lines. GPS measurements at every gravimeter station are required along with visits to the base station at the beginning and end of the survey to account for drift (unless the survey takes multiple days, then return to the base station every few hours and at the start/end of each field day). Teams should utilize both of the CG-5’s available to them, and the survey is estimated to take 5-6 hours.

GPR:

confidence of a survey’s result. The use of gravimeters and GPR aimed to classify a known mine shaft; however, due to limitations in survey designs and a lithology that was not conducive to GPR, we cannot claim uniqueness of our results. With more intensive survey parameters, denser gravimeter coverage, and the addition of electrical resistivity tomography (ERT), the mine shaft structure could be identified and ultimately characterized for public safety uses.

Introduction

Here is the first paragraph of the introduction. The font for this template is Times New Roman, but any comparable font may be substituted. The Section Headings are 9-point bold Times New Roman and the text of the paper is 9-point TimesNew Roman.

Abstracts will be printed on the CD-ROM and online exactly as they are submitted. SEG staff will not edit or retype the copy. The maximum length of an abstract is 4 pages.

Methods

Local variations in subsurface properties affect the way signals propagate through the ground. Mass distribution in the subsurface plays a big role in this effect. If we have a void space, or removal of mass in the subsurface, there will be an interface in which the clay/sandstone meets an air

over the first heading in the NW direction. This survey included two Impulse Radar’s, each with two frequency antennas. The low-frequency impulse radar had 70 MHz and 300 MHz antennas at a 5 cm trigger distance. The higher-frequency impulse radar had 170 MHz and 600MHz antennas at a 2 cm trigger distance.

Figure # shows the mine shaft opening and tunnel going perpendicular to the gravity and GPR survey lines depicted by blue dots and red/purple lines respectively. These survey lines are overlaid on a topographic map, showing the hillside that could be affecting signals.

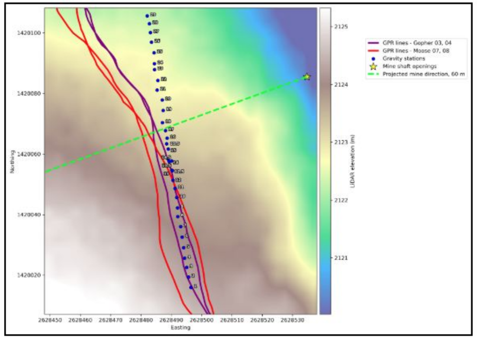


Figure: Map of GPR & Gravity surveys above the historic Onyx Mine atop Howelsen Hill, Steamboat Springs, Co

To Do

- Presentation script
- 25% more of our abstract
- Get presentation feedback & edit accordingly
- Ensure gravity analysis is truly inconclusive :(